

Andrei Baroian

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ABOUT ME

MSc Computer Science (AI) student at Leiden University, currently doing my thesis at [SPY Lab](#) (ETH Zurich) on prompt injection attacks against vision-language models. Past research includes LLM pre-training (architectural variants) and post-training (speeding up GRPO with Prompt Reuse), with smaller projects in quantization and mechanistic interpretability. Excited about autoresearch, scared of cybersecurity capabilities of agents.

PROJECTS & RESEARCH

MSc Thesis: (Image) Prompt Injection

Feb 2026 – Present

[SPY Lab](#), ETH Zurich — Supervised by Jie Zhang and Florian Tramèr. Studying transferability of image prompt injections: adversarial images optimized on open-source VLMs that transfer to closed-source models (GPT, Gemini, Claude). Concurrently exploring prompt injection attacks on [OpenClaw](#) agents, designing self-propagating worm attacks that modify agents' internal goal files and spread to other agents.

Adversarial Attacks on VLA Models in Humanoid Robots

Feb 2026 – Present

[ETH Robotics Club](#) — Robotics Safety Division. Creating adversarial attacks against Vision-Language-Action (VLA) models in humanoid robots. Investigating how visual perturbations can override task instructions and induce harmful behaviors. Concurrently working on defenses.

Prompt Replay: Speeding Up GRPO

[arXiv:2603.21177](#)

An overhead-free online data selection method for GRPO that reuses prompts (not trajectories) to preserve on-policy optimization. Buffers medium-difficulty prompts near a 50% pass rate to maximize learning signal. Tested on Llama-3.2-3B and Qwen3-8B, reducing zero-variance prompts and accelerating early training gains.

Crown, Frame, Reverse: Layer-Wise Scaling Variants for LLM Pre-Training [arXiv:2509.06518](#)

Explored architectural variants that redistribute capacity across transformer layers during pre-training. Introduced three layer-wise scaling patterns using linear interpolation of FFN widths and attention head counts. Pre-trained 180M parameter models on 5B tokens; all variants converged to better performance than an equal-cost isotropic baseline.

Fine-tuning LLMs for Drug Side-Effect Extraction (NER)

[arXiv:2510.22285](#)

Extracted drug-side-effect pairs from patient-forum text. SFT with GPT-4o reached F1 87.1% vs Bio-ClinicalBERT 61.8% and RoBERTa 64.5%. Simplified 2-class setup improved over 5-entity NER.

EDUCATION

Aug 2024 – Jul 2026	MSc Computer Science: Artificial Intelligence , Leiden University, The Netherlands (GPA: 8.5/10) Notable grades: Seminar in Deep Reinforcement Learning (10), Deep Learning (9.0), Seminar in Deep Learning (9.0), Natural Language Processing (9.0).
Feb 2026 – Jul 2026	Exchange Semester — MSc Thesis at SPY Lab , ETH Zurich, Switzerland Adversarial attacks on vision-language models. Supervised by Jie Zhang and Florian Tramèr.
Sep 2021 – Jul 2024	BSc Entrepreneurship & Business Innovation , Tilburg University, The Netherlands

WORK EXPERIENCE

Leiden University - Teaching Assistant, Automated Machine Learning	Sep 2025 – Jan 2026
– Grade assignments and provide feedback. – Guide students in selecting, understanding, and presenting research papers.	
Akida, The Hague - Data Engineer	Mar 2025 – Present
– Develop filtering logic to detect construction projects in public-sector sources using heuristics & GenAI. – Build the extraction pipeline for summarization and structured information retrieval with LLMs. – Collaborate on annotation workflows and quality evaluation of LLMs.	

SKILLS

LLMs	Pre-training (data prep, transformer design, distributed training), post-training (RLVR/GRPO, SFT), prompt engineering, quantization, mechanistic interpretability.
Adversarial ML	Prompt injection attacks (text & image), adversarial patches for VLMs/VLAs, transferability across models.
NLP	Named Entity Recognition, information retrieval, annotation workflows.
Frameworks & Tools	PyTorch; Hugging Face Transformers; Weights & Biases; OpenAI & Google Gemini APIs; Python; Linux; Slurm; Git; Docker; Azure.